

ARAB YOUTH SPACE HACKATHON 2026 • 813 CHALLENGE

Mawrid

Coastal water intelligence for the Banc d'Arguin: where the water is clear enough to monitor from orbit, how the seagrass meadows have changed, and where a boat has to go because the satellite cannot see.

Team	Country	Theme
Malakout	Mauritania	Water Quality & Inland/Coastal Water Intelligence

THE PROBLEM

One of Africa's largest seagrass systems, and nobody is measuring it

12,000

km² of shallows in the Parc National du Banc d'Arguin, a UNESCO World Heritage site

2M+

migratory shorebirds depend on these flats; the national fishery depends on the nursery they shelter

0

continuous measurements of meadow extent. Field campaigns cover a few transects, years apart

Who is affected. Seagrass systems elsewhere have collapsed in years, not decades, and marine heatwaves have driven losses of comparable meadows in Europe. If that is starting here, the people who would feel it first are the artisanal fishers of the park and the Imraguen communities inside it — and today nothing would detect it until a campaign happened to find it.

THE BUSINESS USE CASE

We do not sell a map. We sell where to send the boat.

- **The end user** is the management authority of the Parc National du Banc d'Arguin, and the institutes and NGOs that advise it.
- **The decision** is where to spend a handful of boat days per season across 12,000 km².
- **Today** that decision rests on expert judgement, because no basin-wide layer says where something changed.

PRODUCT	ANSWERS
Corrected series	Has it changed, by how much, and is that above the noise?
Observability layer	Which parts of the basin can a satellite measure at all?
Priority lists	Given a fixed budget, where does the next survey go?

11 ranked field sites and 20 sonar targets, each with coordinates and a stated reason — a route instead of a season of guessing.

Sentinel-2, and one piece of luck

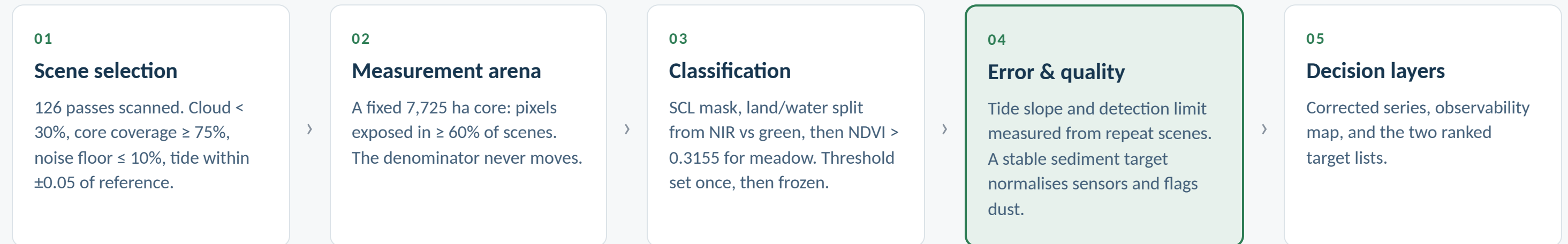
FIELD	VALUE
Product	Sentinel-2 Level-2A surface reflectance
Provider	ESA Copernicus · Earth Search STAC
Bands	B02, B03, B04, B08, SCL
Period	2017–2026 · 72 scenes used
Resolution	10 m, EPSG:32628
Licence	Open and free

The luck. Turbid water defeats optical remote sensing in this basin — we measured exactly how badly. But twice a day the tide takes the water off these meadows and hands the satellite a clear view of its own intertidal zone, free, every five days, since 2017.

Hyperspectral. We use none, and we will not pretend otherwise. Narrower bands would separate seagrass from green algae and wet sediment, which is a real limit of a single NDVI threshold. They would **not** solve turbidity: no band recovers a bottom signal that never reached the sensor. We would adopt it for species separation, not for water clarity.

APPROACH

Every threshold frozen before the series was run

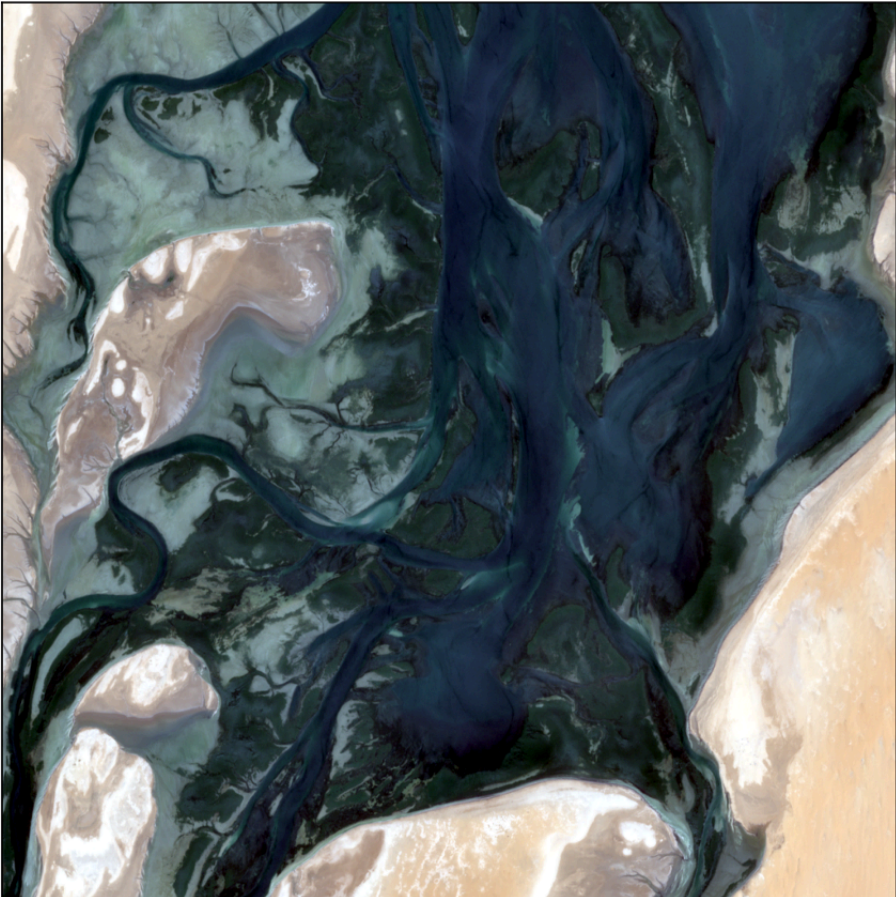


Step 04 is the project. Anyone can threshold an index. What makes this a measurement is that the error bar was measured from the data, and that years which failed the quality rules were dropped rather than patched.

What the classifier sees

Mawrid - S2B 2026-06-05 - Tidra, Banc d'Arguin - exposure index 0.67 - 1,919 ha meadow

True colour



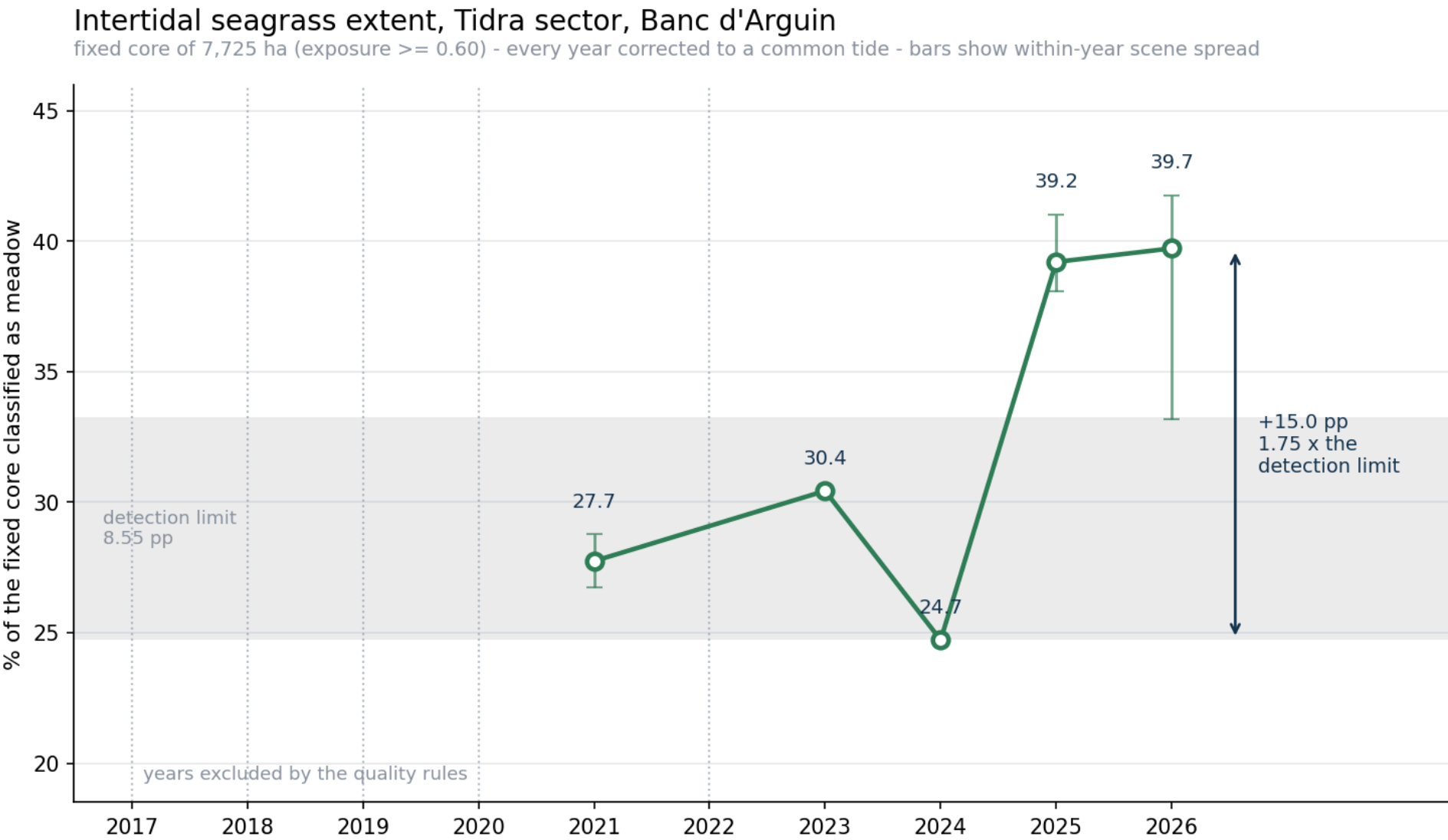
Classification



- Submerged
- Exposed bare sediment
- Exposed meadow
- Masked

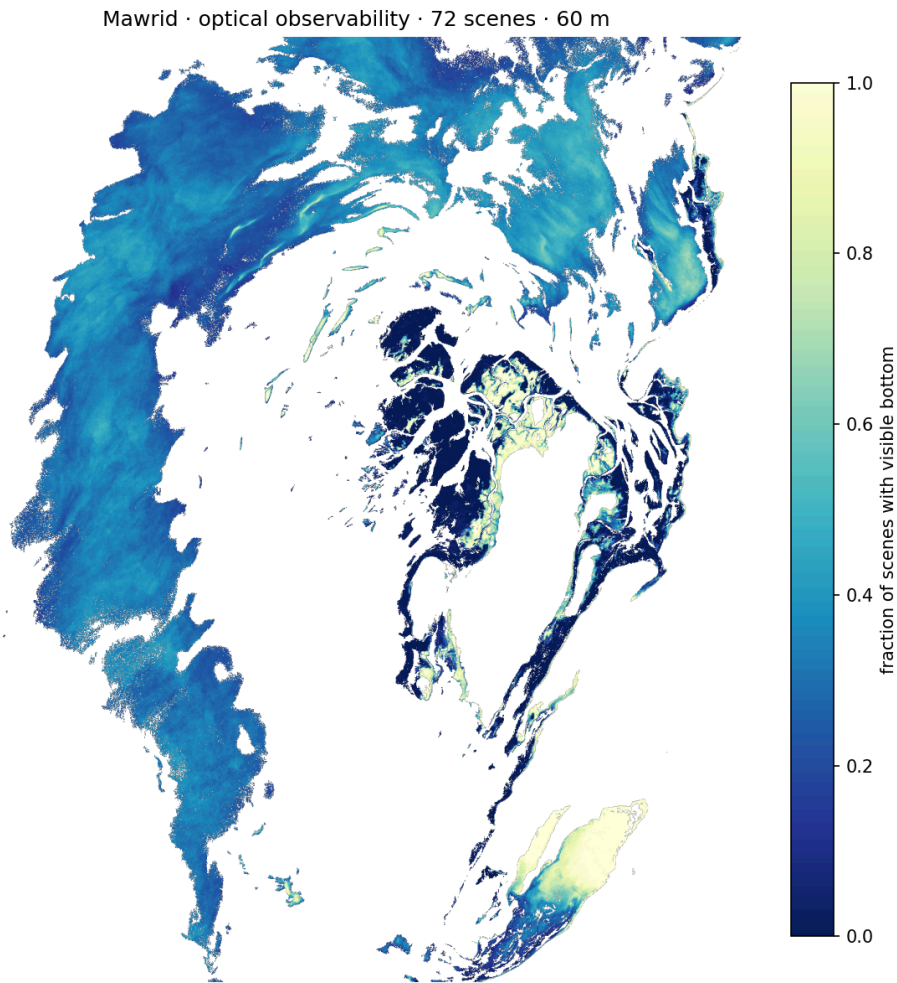
S2B, 5 June 2026, Tidra sector. Exposure index 0.67 — two thirds of the valid pixels were out of the water. 1,919 ha classified as meadow in this 10.24 km window. Reproduced by the notebook in the repository in under a minute.

A rise of 15 percentage points, against a measured limit of 8.55



Five usable years in a fixed 7,725 ha core, each corrected onto a common tide. Dotted lines are years excluded by the quality rules, each for a stated numeric reason. The grey band is the detection limit, measured from repeat scenes of the same unchanged meadow.

Where the satellite is blind



6.3%

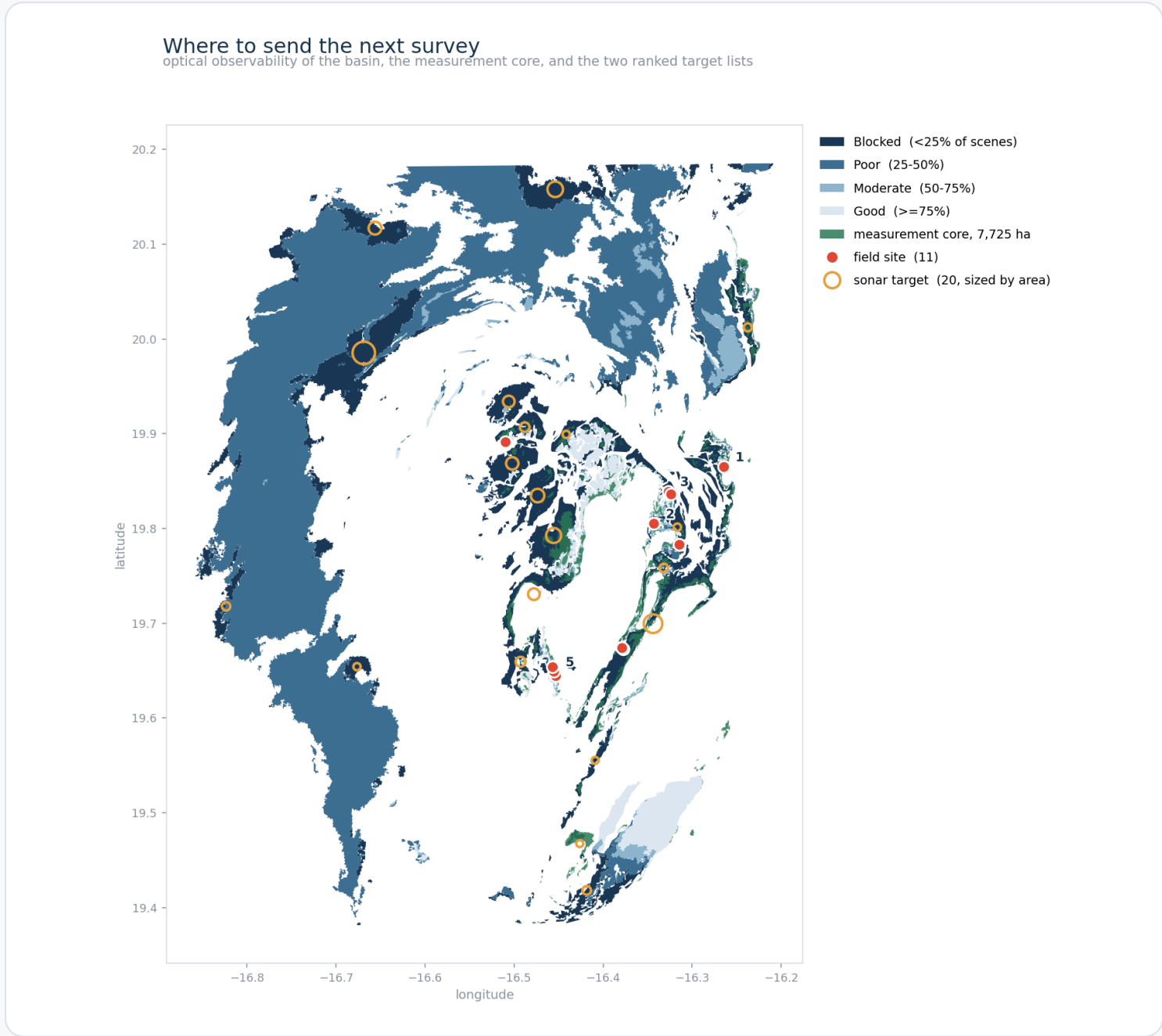
of the submerged area has its bottom visible in 75% or more of 72 scenes — the only part an optical satellite can monitor with confidence

24.7%

is effectively blind: the bottom is visible in under a quarter of scenes. 44,945 ha that no optical method will ever report on

We built a submerged-meadow branch and withdrew it. This layer is what replaced it — and it is the more useful product, because an area the satellite cannot resolve is an area that needs a boat.

The deliverable: a route, not a map



- **11 field sites**, ranked by classification uncertainty × change magnitude × observability. Places where the satellite disagrees with itself and can still be checked from shore.
- **20 sonar targets**, the blind components above 50 ha. The largest is 5,158 ha at 19% observability — no optical sensor will ever answer for it.
- Every entry carries coordinates, area, and the reason it is on the list.

This is the layer a park authority can act on tomorrow, and the one a survey contractor would pay for.

VALIDATION

The error bar was measured, not assumed

-81.4

percentage points per unit of exposure index. The tide sensitivity, estimated from repeat scenes **within** the same year — so it cannot manufacture the between-year trend it is used to test

8.55

percentage points. Two scenes of the same year, same unchanged meadow, disagree by this much. That is the detection limit, and nothing smaller is reported as a change

1.75×

the detection limit. The 2024→2026 rise of 15.0 pp clears it — which makes it a likely trend, and not yet a proven one

Quality rules, applied without exception. A stable bare-sediment target is classified in every scene; the share of it falsely called meadow is the noise floor. Two years failed on that test alone — one on a sensor offset, one on a Saharan dust signature. Five of ten years are missing, each with a numeric reason recorded in `results/series.json`.

WHAT WE GOT WRONG

We retracted our own headline result

The claim we made first

A loss of seagrass between two dates, measured by differencing two classified scenes. It was clear, it was alarming, and it was the centre of our first submission draft.

Why it did not survive

A marine ecologist who works on this system pointed out that the two dates were taken at different tides. We built the test. The tide alone explains a change of that size. The claim was withdrawn.

Everything in this deck exists because of that failure. The fixed core, the tide correction, the detection limit, the quality rules and the observability layer were all built to make sure the next claim could not fail the same way. A proof of concept that cannot survive its own strongest critic is not worth incubating.

LIMITATIONS

Where this breaks, stated plainly

- **Intertidal only.** Everything reported here concerns the core that is exposed at low water. The deep meadows are beyond this method — that is what the sonar list is for.
- **Five of ten years are missing.** 2017, 2018, 2019, 2020 and 2022 failed the quality rules. The series is five points, not a continuous record, and it is not interpolated.
- **No field validation.** Nothing has been checked against a transect. "Likely trend" is the strongest statement the evidence supports.
- **One index, one threshold.** NDVI does not separate seagrass from green algae or from wet sediment at the margin. This is where hyperspectral would genuinely help.
- **The quality rules were tightened after the first series run.** We state that in the repository rather than presenting them as having been fixed from the start.

IMPACT

What changes if this runs every year

For the park

A basin-wide change signal with a known error bar, refreshed every season instead of every few years — and a survey plan that follows it.

For the fishery

The nursery habitat behind the national catch becomes something that can be monitored, and a decline becomes something that can be seen early enough to act on.

For the region

Every step is open data and open code. The same pipeline runs on any intertidal coast in the Arab world with a tidal range and a Sentinel-2 archive.

Aligned with **SDG 14** (life below water) and **SDG 13** (climate action): intertidal seagrass is both a habitat indicator and a carbon-storing system, and this is the measurement layer either argument needs before it can be made honestly.

NEXT STEPS • INCUBATION

From a likely trend to a measured one

01

Field validation

Transects on the 11 ranked sites, to turn the detection limit into a validated accuracy

02

Acoustic survey

The largest blind components, starting with the 5,158 ha target

03

Full park coverage

Extend from the Tidra sector to the whole of the Banc d'Arguin

04

Species separation

Hyperspectral bands, for the one limit they can actually fix

Team Malakout • Mauritania • Theme: Water Quality & Inland/Coastal Water Intelligence

Code, data and notebook: github.com/El-Arby571/mawrid